



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to T. RANGAPPAN & SONS UNIT II

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

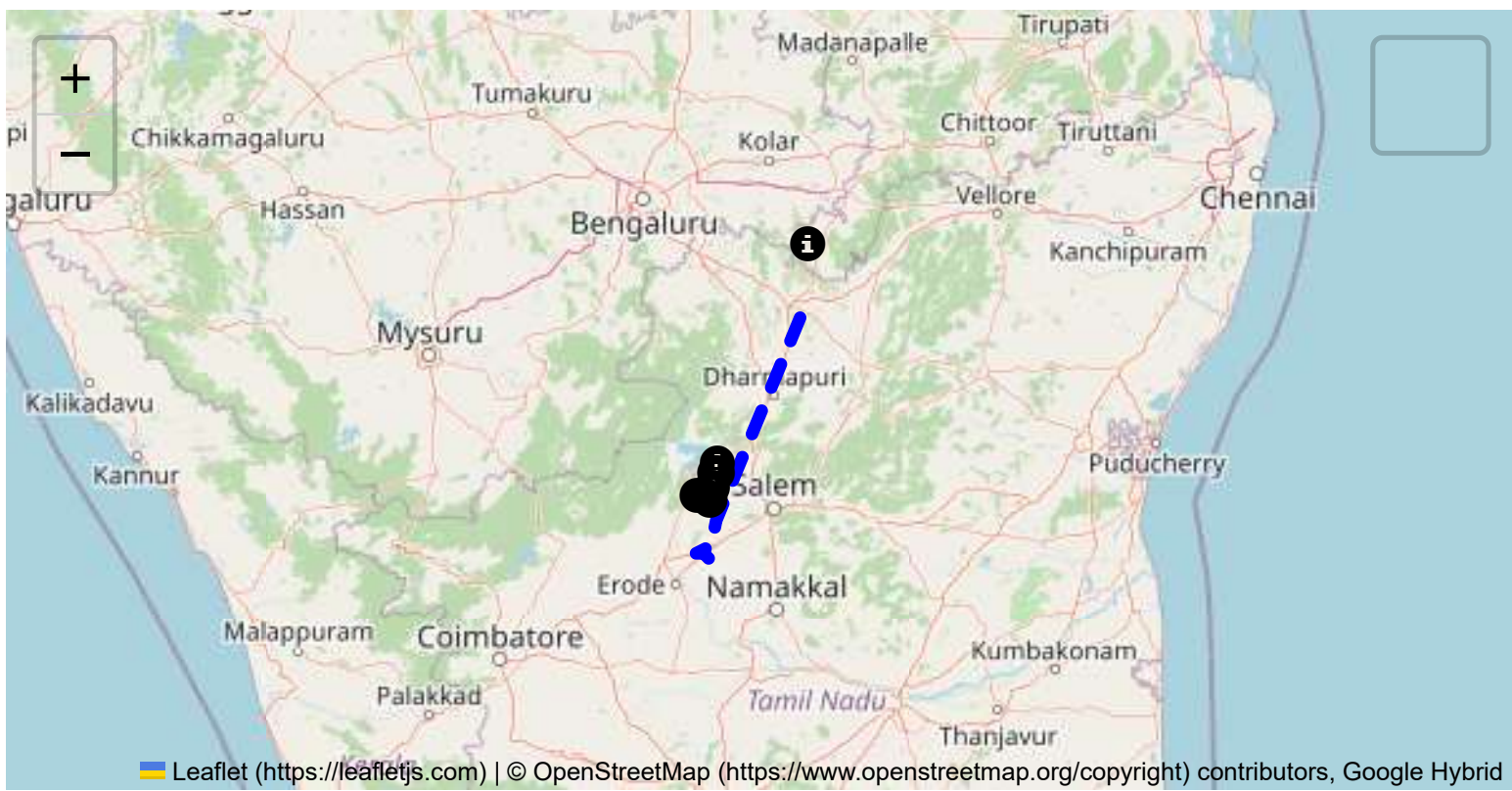
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

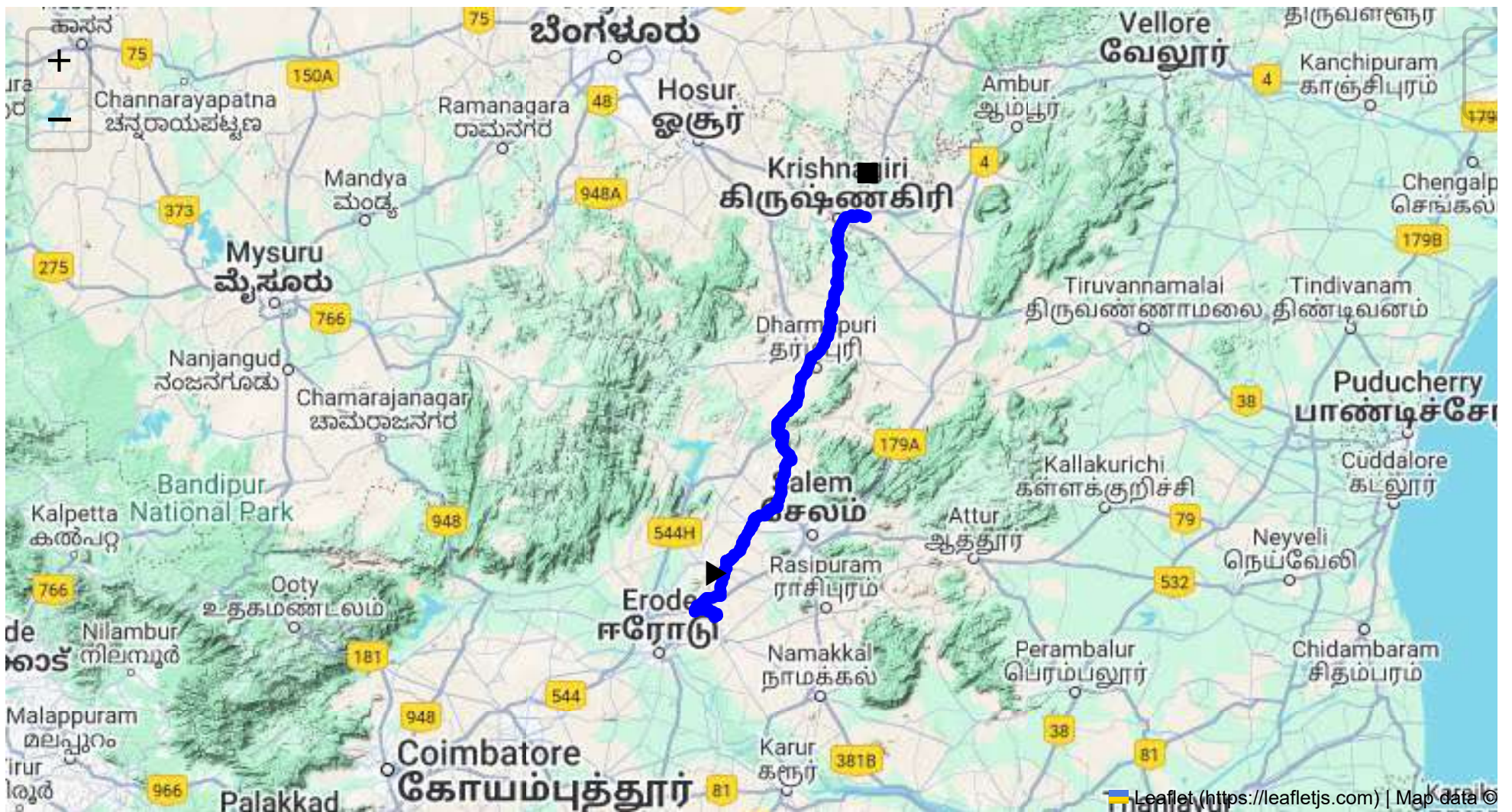
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 159.12 km
Estimated Duration: 3.0 hours
Adjusted Duration (Heavy Vehicle): 3.8 hours
Start: (11.4381, 77.8734)
End: (12.534722, 78.294948)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from CVQF+23W, Sangagiri, Tamil Nadu to G7MW+V44, Balinayanapalli, Tamil Nadu is primarily along local roads such as Sankari RS - Erode Privu Rd and Bhavani Main Road, continuing on SH 86. The route covers a variety of rural and semi-urban landscapes, connecting several small towns and villages.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu experiences tropical weather with hot summers, a monsoon season typically from October to December, and mild winters. The monsoon can result in heavy rainfall, leading to waterlogging and slippery roads, which could be hazardous for heavy vehicles. Fog can occasionally be an issue in winter months, reducing visibility.

3. Traffic Patterns and Congestion Areas: Traffic can vary significantly, with morning and evening times typically being peak hours, especially near towns like Sankari and Erumaipatti. Sankari RS - Erode Privu Rd and Bhavani Main Road in particular may experience congestion during these periods due to local commuter traffic. Markets or festive occasions can lead to additional traffic.

4. Road Quality and Infrastructure: While main highways such as SH 86 may be relatively well-maintained, local roads, particularly less-trafficked ones, can suffer from potholes and uneven surfaces. Infrastructure such as lighting and signage may be insufficient in rural areas, particularly during nighttime driving.

5. Suggestions for Alternative Routes for Emergencies: In case of emergencies, it is advisable to use NH 44, a major highway in the vicinity, as it offers better connectivity and facilities. This could be accessed through nearby towns depending on the location along the route.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Tamil Nadu follows standard Indian regulations for the transport of hazardous materials, requiring appropriate permits, vehicle signage, and documentation. Trucks should ensure compliance with weight limits and driving hours to avoid penalties.

7. Historical Incidents Involving Heavy Vehicles or Hazardous Materials: Information on historical incidents could be limited but it is important to be cautious in curves and steep inclines, which can be dangerous for heavy vehicles. Local news archives or law enforcement agencies might provide specific details on accident-prone zones.

8. Environmental Considerations and Sensitive Areas: The route passes through small towns and some agricultural areas; hence, spillage or accidents involving hazardous materials could severely impact local environments. Transporters should exercise extreme caution in handling materials.

9. Analysis of Communication Coverage: While communication networks have improved, rural sections of the route may experience spotty coverage, particularly in more isolated regions between smaller villages. Drivers should have physical maps and emergency contacts ready.

10. Estimated Emergency Response Times: Emergency response can vary widely along the route: towns may have relatively quick access to services, while rural stretches might face delays due to distance from facilities. Averagely, expect between 30 minutes to 1 hour depending on proximity to larger towns.

11. Overall Summary of Risk Assessment: The route poses moderate risk due to road quality, occasional traffic congestion, and potential weather-related issues during the monsoon. It requires vigilant adherence to driving safety, particularly regarding speed management and maintaining safe distances. Drivers should be prepared for extended driving times owing to congestion or road conditions and always prioritize communication with local authorities in case of any emergency situation. Due diligence in route planning and adherence to regional regulations on hazardous materials are crucial for ensuring safety on this journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	High	11.45615, 77.81677	15 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	Medium	11.64171, 77.95184	30 KM/Hr
16	Turn	Medium	11.64243, 77.95198	30 KM/Hr
17	Turn	Medium	11.64721, 77.95118	30 KM/Hr
18	Turn	Medium	11.67870, 77.96846	30 KM/Hr
19	Turn	High	11.70227, 77.97864	15 KM/Hr
20	Turn	Medium	11.71416, 77.99604	30 KM/Hr
21	Turn	Medium	11.71715, 77.99969	30 KM/Hr
22	Turn	Medium	11.71921, 78.01133	30 KM/Hr
23	Turn	Medium	11.72737, 78.02248	30 KM/Hr
24	Turn	Medium	11.73775, 78.03465	30 KM/Hr
25	Turn	High	11.73651, 78.03878	15 KM/Hr
26	Turn	Medium	11.74189, 78.04660	30 KM/Hr
27	Turn	Medium	11.74309, 78.04725	30 KM/Hr
28	Turn	Medium	11.77961, 78.05775	30 KM/Hr
29	Turn	Medium	11.86873, 78.08374	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	Medium	11.87966, 78.08187	30 KM/Hr
31	Turn	Medium	11.92609, 78.06513	30 KM/Hr
32	Turn	Medium	11.93719, 78.05372	30 KM/Hr
33	Turn	Medium	11.95457, 78.04685	30 KM/Hr
34	Turn	Medium	11.95688, 78.04803	30 KM/Hr
35	Turn	High	11.95828, 78.04983	15 KM/Hr
36	Turn	Medium	11.95997, 78.04946	30 KM/Hr
37	Turn	Medium	11.96335, 78.05141	30 KM/Hr
38	Turn	High	11.96408, 78.05131	15 KM/Hr
39	Turn	Medium	11.96511, 78.04966	30 KM/Hr
40	Turn	High	11.96608, 78.04919	15 KM/Hr
41	Turn	Medium	11.96819, 78.05088	30 KM/Hr
42	Turn	Medium	12.17005, 78.16792	30 KM/Hr
43	Turn	Medium	12.19919, 78.18172	30 KM/Hr
44	Turn	Medium	12.22472, 78.19140	30 KM/Hr
45	Turn	Medium	12.24623, 78.19524	30 KM/Hr
46	Turn	Medium	12.25078, 78.19988	30 KM/Hr
47	Turn	Medium	12.25370, 78.20034	30 KM/Hr
48	Turn	Medium	12.29150, 78.19930	30 KM/Hr
49	Turn	Medium	12.40604, 78.22285	30 KM/Hr
50	Turn	Medium	12.42500, 78.22672	30 KM/Hr
51	Turn	Medium	12.50188, 78.22338	30 KM/Hr
52	Turn	High	12.50722, 78.22222	15 KM/Hr
53	Turn	Medium	12.53182, 78.23982	30 KM/Hr
54	Turn	Medium	12.53345, 78.28926	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
35	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
29	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
31	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium
32	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
33	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
34	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
36	college	Government Boys Arts and Science college	12.5323457, 78.2457274	30 km/h	Medium
37	college	Government Polytechnic College	12.5352064, 78.2460463	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59214, 77.90656



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71416, 77.99604



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74189, 78.04660



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74309, 78.04725



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77961, 78.05775



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86873, 78.08374



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87966, 78.08187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92609, 78.06513



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93719, 78.05372



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95457, 78.04685



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95688, 78.04803



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.95828, 78.04983



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95997, 78.04946



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96335, 78.05141



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96408, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96608, 78.04919



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96819, 78.05088



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17005, 78.16792



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.19919, 78.18172



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22472, 78.19140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24623, 78.19524



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25078, 78.19988



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25370, 78.20034



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29150, 78.19930



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40604, 78.22285



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.42500, 78.22672



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.50188, 78.22338



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.50722, 78.22222



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.53182, 78.23982



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.53345, 78.28926
